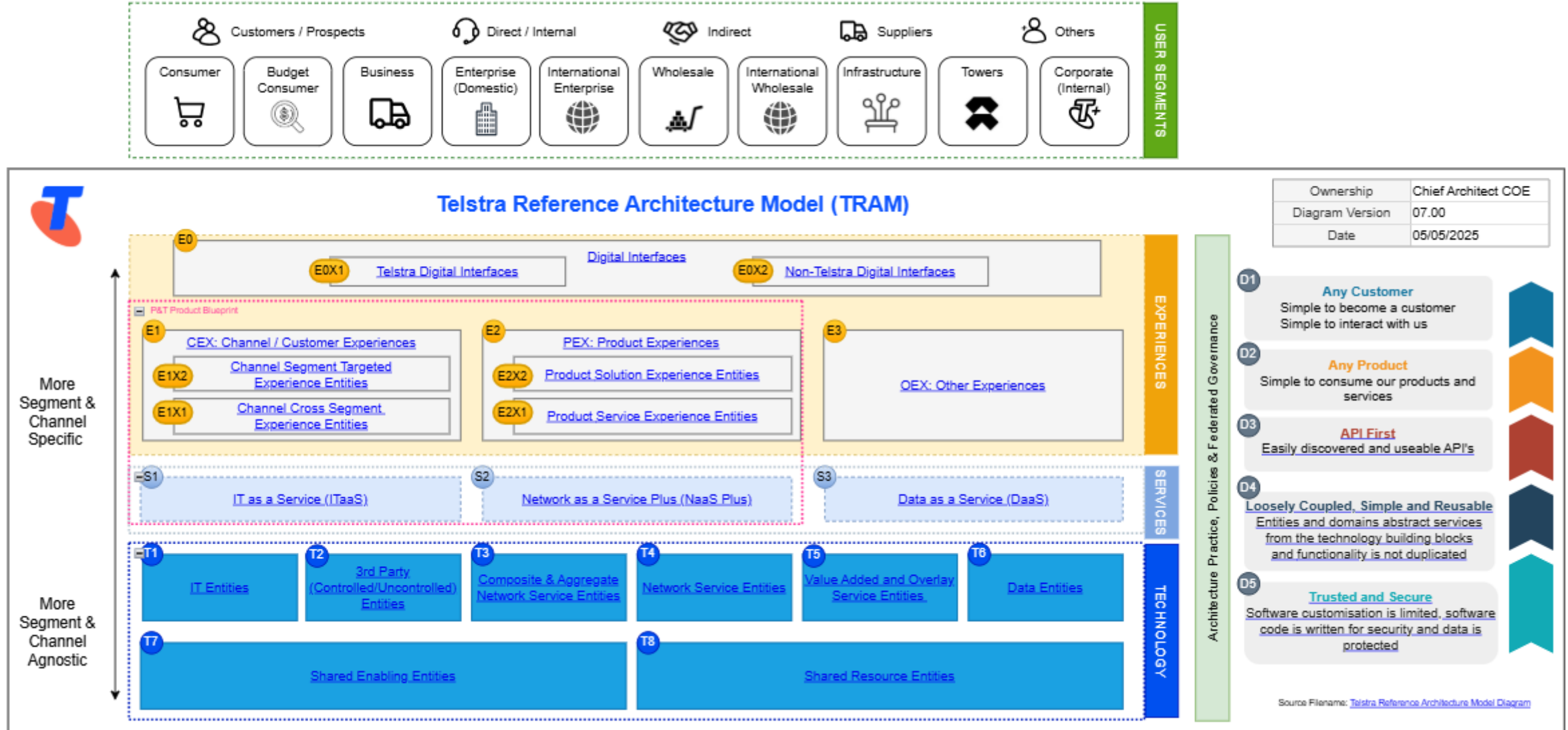


Operational Data Store usecase and required proof points.



ODS in TRAM as a T7



ODS in TRAM as a T7



Shared Enabling Entity - Component Technologies (SEE-CT)

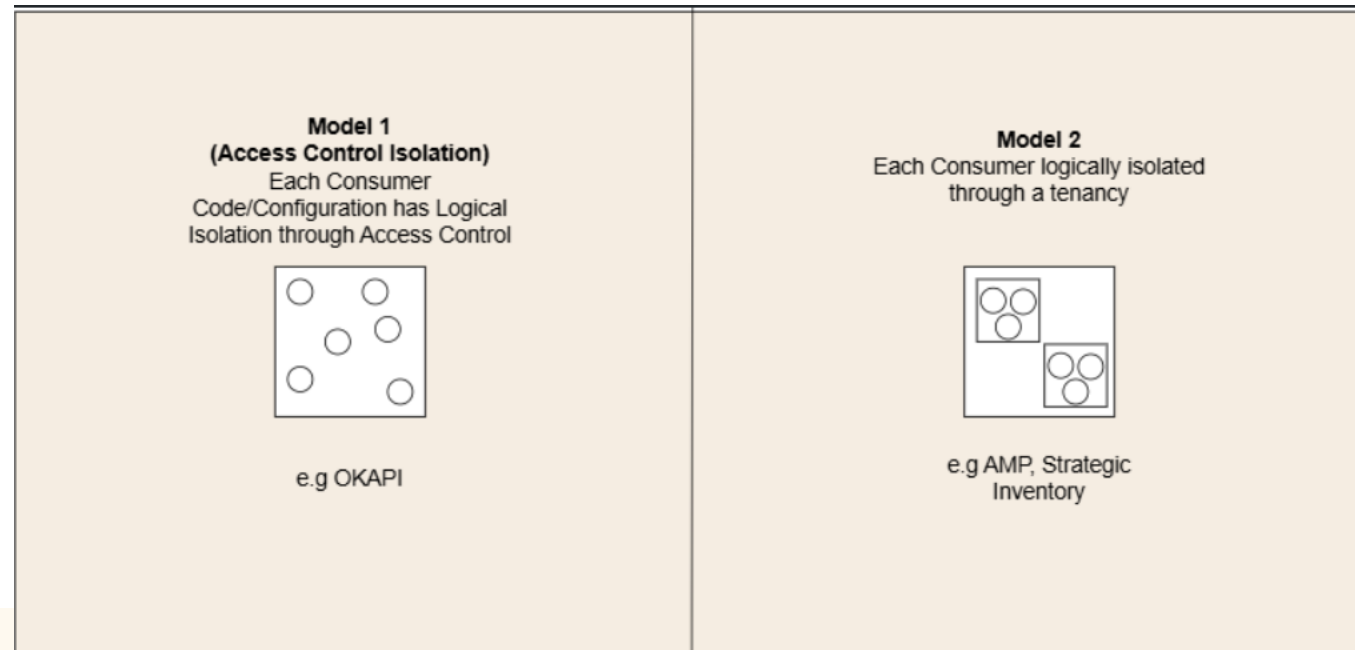
IaaS, PaaS and SaaS are the three most popular types of Shared Enabling Entities offered to Consumer Entity without abstraction:

- IaaS, or Infrastructure as a service, is on-demand access to physical and virtual servers, storage and networking - the backend infrastructure for running workloads of Consumer Entities.
- PaaS, or Platform as a service, is on-demand access to complete, read-to-use platform for developing, running, maintaining and managing applications of Consumer Entities.
- SaaS, or Software as a service, is on-demand access to ready-to-use, application software as required by Consumer Entities.

SEE-CT offers underlying technology which is visible to the Consumer entities, and there is a tight coupling between Consumer Entity's functions and technology offered by Shared Enabling Entity. A change in vendor or technology of SEE-CT will impact Consumer Entity and may incur costs of migration.

	IaaS	PaaS	SaaS
Code & Configuration			
Applications			
Data			
Runtime			
Middleware			
OS			
Virtualisation			
Server			
Storage			
Networking			

	Managed by Consumer Entity		Managed by Shared Enabling Entity
--	----------------------------	--	-----------------------------------



ODS Functional Requirements



Functional Requirements

- FR-1: Multi-Tenant ODS Capability
- FR-2: Tenant Lifecycle Management
- FR-3: Real-Time and Near-Real-Time Data Ingestion
- FR-4: Support for Heterogeneous Data Domains
- FR-5: Schema-Flexible Data Storage
- FR-6: Operational Query Capability
- FR-7: Reference Data Enrichment
- FR-8: Topology-Aware Data Access
- FR-9: API-Based Data Access
- FR-10: Support for Industry and Telstra-Specific APIs
- FR-11: Operational Feature Enablement
- FR-12: Integration with Operational Workflows
- FR-13: Controlled Access to Operational Data
- FR-14: Separation of Operational and Historical Scope

ODS Functional Requirements



FR-1: Multi-Tenant ODS Capability

The ODS must operate as a shared T7 capability that supports multiple independent tenants within a single platform instance, enabling teams to consume ODS functionality without deploying dedicated infrastructure.

FR-2: Tenant Lifecycle Management

The ODS must provide tenant lifecycle functionality, including tenant creation and onboarding, tenant configuration and activation, and tenant suspension and archiving. This capability must be exposed in a way that enables tenant teams to self-manage their presence on the platform.

FR-3: Real-Time and Near-Real-Time Data Ingestion

The ODS must ingest operational data streams from upstream systems, including real-time telemetry, near-real-time network and service metrics, and operational metadata required to contextualize telemetry. The ingestion capability must support continuous, streaming-oriented data flows consistent with operational use cases.

FR-4: Support for Heterogeneous Data Domains

The ODS must store and manage multiple operational data domains, including network telemetry, service performance metrics, topology data, and customer and service reference metadata. These domains must coexist within the ODS to support operational correlation and enrichment.

FR-5: Schema-Flexible Data Storage

The ODS must support flexible and evolving data schemas, allowing new metrics, attributes, and structures to be introduced without requiring redesign of existing consumers. This includes support for nested and hierarchical data structures commonly found in topology and telemetry datasets.

FR-6: Operational Query Capability

The ODS must provide query capabilities suitable for operational use, enabling consumers to retrieve current and recent operational state, query service-level and component-level data, and filter and project data based on operational attributes. Queries must be usable by both internal systems and exposed APIs.

FR-7: Reference Data Enrichment

The ODS must support enrichment of operational data using reference and topology data, including mapping Customer-Facing Services (CFS) to Resource-Facing Services (RFS) and resolving service identifiers to network identifiers. This enrichment capability is required to bridge customer and network perspectives.

FR-8: Topology-Aware Data Access

The ODS must support topology-aware data access, enabling consumers to traverse service hierarchies, associate telemetry with composite and atomic service components, and aggregate operational data across service structures.

ODS Functional Requirements



FR-9: API-Based Data Access

The ODS must expose data access via APIs, supporting internal operational consumers, telemetry and service performance APIs, and integration with downstream analytics, assurance, and automation systems. The API layer must be extensible to support multiple API standards.

FR-10: Support for Industry and Telstra-Specific APIs

The ODS must support implementation of industry and Telstra-specific APIs, including TMF-aligned Service Performance APIs and internal telemetry and operational APIs. The ODS must act as a backend system of record for these APIs.

FR-11: Operational Feature Enablement

The ODS must support the derivation and storage of operational features, including derived metrics, rolling aggregates, and enriched operational attributes. These features must be accessible to downstream systems such as anomaly detection and assurance workflows.

FR-12: Integration with Operational Workflows

The ODS must integrate with operational systems and workflows, enabling use cases such as anomaly detection, service assurance, operational alerting, and customer experience insights. The ODS is explicitly positioned as an operational, not purely analytical, data store.

FR-13: Controlled Access to Operational Data

The ODS must support controlled access to operational data, enabling internal system consumers, robot or service consumers, and (where applicable) customer-scoped access via upstream authorization mechanisms.

FR-14: Separation of Operational and Historical Scope

The ODS must focus on current and near-term operational data, with long-term historical storage and analytics explicitly out of scope. Where historical data is required by APIs, the ODS must provide integration points rather than acting as the system of record for history.

ODS Non-Functional Requirements



Non-Functional Requirements

Preface

NFR-1: Tenancy Isolation

NFR-2: Tenant Archiving

NFR-3: Self-Service Enablement

NFR-4: API Layer Modularity

NFR-5: Availability

NFR-6: Geographic Redundancy

NFR-7: Consistency Model

NFR-8: Query Concurrency

NFR-9: Indexing and Query Performance

NFR-10: Topology and Enrichment Scalability

NFR-11: Access Control Flexibility

NFR-12: Operational vs Analytical Workload Isolation

NFR-13: Observability

ODS Non-Functional Requirements



NFR-1: Tenancy Isolation

The ODS must ensure strict logical isolation between tenants, including data, metadata, indexes, and access policies, such that no tenant can access or infer another tenant's data.

NFR-2: Tenant Archiving

The ODS must support tenant archiving, enabling tenants to be removed from active ingestion and compute paths while preserving data and metadata for compliance or potential reactivation.

NFR-3: Self-Service Enablement

The ODS must enable tenant teams to self-serve onboarding and runtime configuration without requiring platform operator intervention.

NFR-4: API Layer Modularity

The ODS API layer must be modularly decoupled from core data storage and query components, such that changes or extensions to API standards do not require modification of core data models or query execution logic.

NFR-5: Availability

The ODS must achieve ultra-high availability, including during maintenance activities such as patching and upgrades.

NFR-6: Geographic Redundancy

The ODS must support geographic redundancy using a clearly defined deployment model (e.g. active/active or active/standby).

NFR-7: Consistency Model

The chosen geo-redundancy model must explicitly define data consistency guarantees, including what data is strongly consistent, what data may be eventually consistent, and how consumers are protected from stale or partial data.

NFR-8: Query Concurrency

The ODS must support multiple concurrent long-running read and write queries without causing service degradation or starvation of operational workloads.

NFR-9: Indexing and Query Performance

The ODS must support flexible indexing strategies and query execution over nested and evolving data structures without requiring disruptive re-indexing or data reloads.

NFR-10: Topology and Enrichment Scalability

Topology traversal and enrichment operations must scale with increasing service hierarchy depth and data volume without requiring full pre-materialisation.

NFR-11: Access Control Flexibility

The ODS must support flexible authorisation models, including RBAC and ABAC, and enable delegated authorisation via upstream systems where required.

NFR-12: Operational vs Analytical Workload Isolation

Operational workloads must be protected from degradation caused by analytical or exploratory usage of the ODS.

NFR-13: Observability

The ODS must expose platform and tenant-level observability metrics, including ingestion lag, query latency, error rates, and availability.

ODS Functional Requirements

Aspect	Operational NFR's (Systems)
Hosting	Can be hosted in On-Premise and Public Cloud (e.g. GCP, AWS)
Security & Compliance	Infrastructure level (encryption at rest/in-transit, VPC, IRAP, zero trust)
Latency	Sub-second to seconds (e.g. <100 ms for ODS queries, <500 ms for inference) ; throughput initial 200k eps, scalable to 1M eps
Availability	99.9 to 99.99% (often 24x7 with zero-downtime deployments) Infrastructure as Code; blue/green or canary for job updates.
Concurrency	Very high (thousands of concurrent sessions/queries)
Scalability	Horizontal scaling of compute + storage independently (cloud-native) storage tiers with retention policies
Recovery Time Objectives (RTO)	Minutes (hot-hot or active-active setups) $\leq$ 1 hour.
Recovery Point Objectives (RPO)	Near-zero (Realtime replication) RPO $\leq$ 15 minutes
Data Freshness	Near real-time / real-time (seconds)
Retention period in ODS data store	18 months
Data Accuracy / Quality	$\geq$ 99.5% events conform to schema; $\leq$ 0.1% duplication at sink technical accuracy, consistency enforced at source
Schema	Schema-on-write or schema-less/flexible (modern ODS)
Observability/Telemetry	Infrastructure metrics, traces, logs (SPLUNK, Prometheus, Confluent control center)
Key examples	• ODS serving 10,000 qps (queries per second) • Kafka cluster with 99.99% uptime • Edge inference latency < 50ms • Current Observability that will be in ODS is currently writing close to 500k events/sec today. Expecting the event rate to reach several million per second over 2 years.

ODS high level

